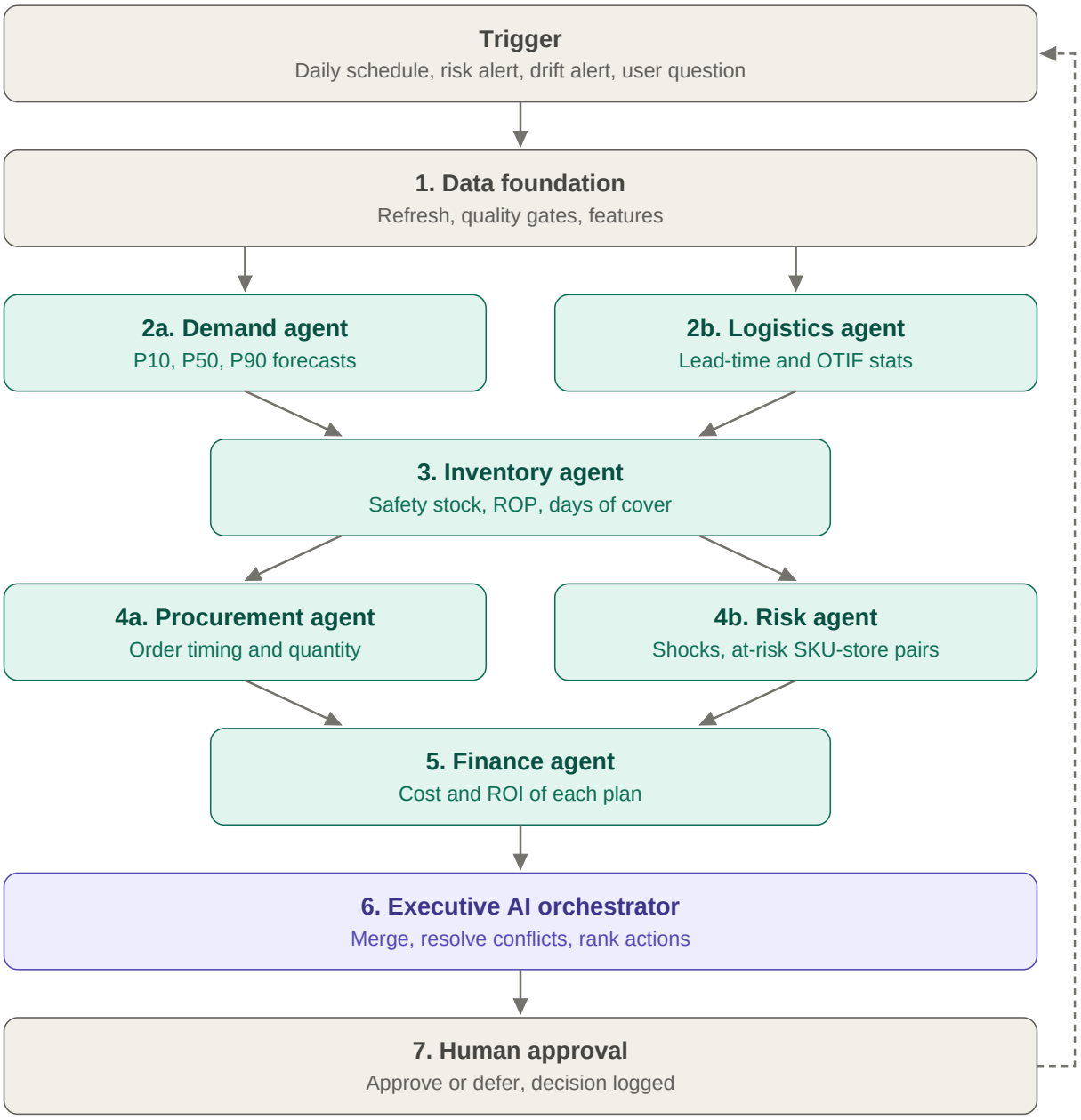
Agent Control Flow

One control tower cycle: how the six agents and the orchestrator hand work to each other



Numbers show run order. Steps 2a/2b and 4a/4b run in parallel. The dashed line is the feedback loop: approved actions become the new stock position for the next cycle, and alerts can trigger an early run.

How the six agents work together

The six agents are Demand, Inventory, Procurement, Logistics, Risk and Finance, with the Executive AI orchestrator on top.

How they talk

The six agents are not six chatbots messaging each other. Each is an analytical module (a forecasting model, an optimizer, a simulator) with a fixed input and output format. They pass structured data through a shared state store, and the orchestrator decides who runs when. Agents do not call each other freely, so every number can be traced to one agent. This is also why the orchestrator never calculates anything itself: it reads agent outputs, ranks them, and explains.

What each agent hands to the next

From	To	What is passed
Demand	Inventory, Risk	P10/P50/P90 forecasts per SKU-store-day, plus the drivers used (promo, weather, epidemic flag)
Logistics	Inventory, Procurement, Risk	Lead-time average and variability, OTIF and fill rate per supplier
Inventory	Procurement, Risk, Finance	Safety stock, reorder point, projected days of cover, reorder triggers
Procurement	Finance, Orchestrator	Proposed order timing and quantity, rounded to MOQ
Risk	Orchestrator, Procurement	At-risk SKU-store pairs, shock severity, regime flag; can request an expedite
Finance	Orchestrator	Cost and ROI of each candidate plan, flagging plans where cost exceeds benefit

Parallel steps and conflicts

- **Parallel steps.** Demand and Logistics run side by side because neither needs the other. Procurement and Risk also run side by side once Inventory has published its policy. Finance waits until both have produced plans.
- **Conflicts.** If Risk says "expedite this order" and Finance says "the expedite costs more than the stockout risk", the orchestrator does not pick a winner. It shows the human both numbers and the trade-off, and the decision is logged.

Event-driven loops

- **Normal loop.** Approved actions become the new stock position for the next cycle.
- **Shock alert.** If Risk detects a shock, such as an epidemic flag, the orchestrator re-runs Demand with that regime switched on, and everything downstream follows.
- **Drift alert.** A drift alert from monitoring triggers a Demand retrain.

Limits from the data

Logistics works at supplier level only, since no dataset has shipment tracking. Finance relies on Target's cost fields, so Atomicd-based scenarios use Target's cost parameters as an assumption.

Prototype versus full build

In the capstone build, the agents can be plain Python modules wrapped as tools in LangChain or CrewAI, and only the orchestrator and copilot use an LLM. The prototype's copilot tab uses scripted answers for this reason.